

CS 506: An Introduction to Quantum Computing

Loose Analysis of Grover's Search Algorithm

Scribe: Neil P Thimmaiah

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Loose Analysis of Grover's Search Algorithm

We are given a function $f: \{0, 1, 2, \dots, N-1\} \rightarrow \{0, 1\}$ such that

- (a) There exists a unique marked element x^* such that: $f(x^*)=1$
- (b) All other $f(x) = 0$

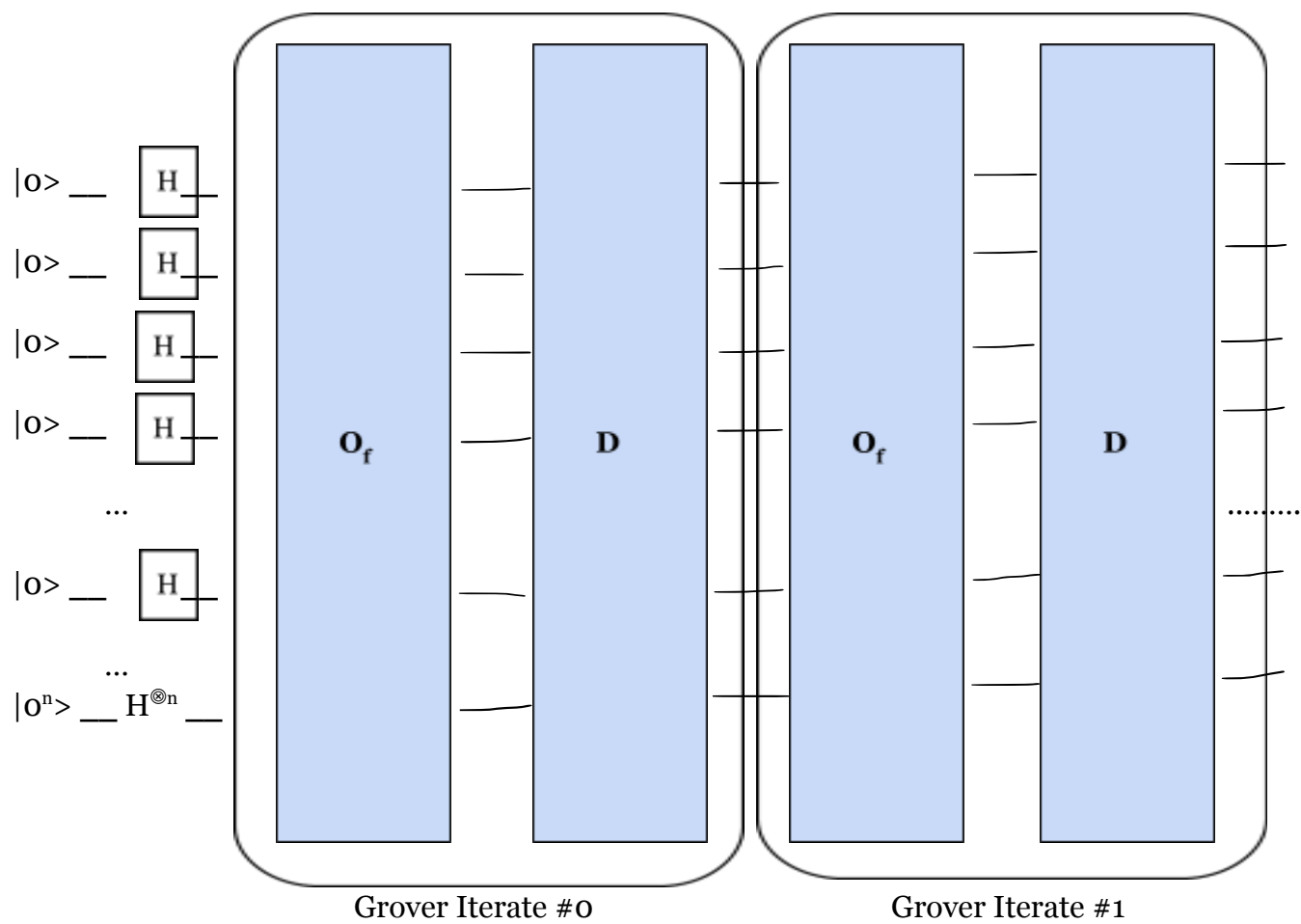
Goal: Find x^* with probability $\geq 1/100$

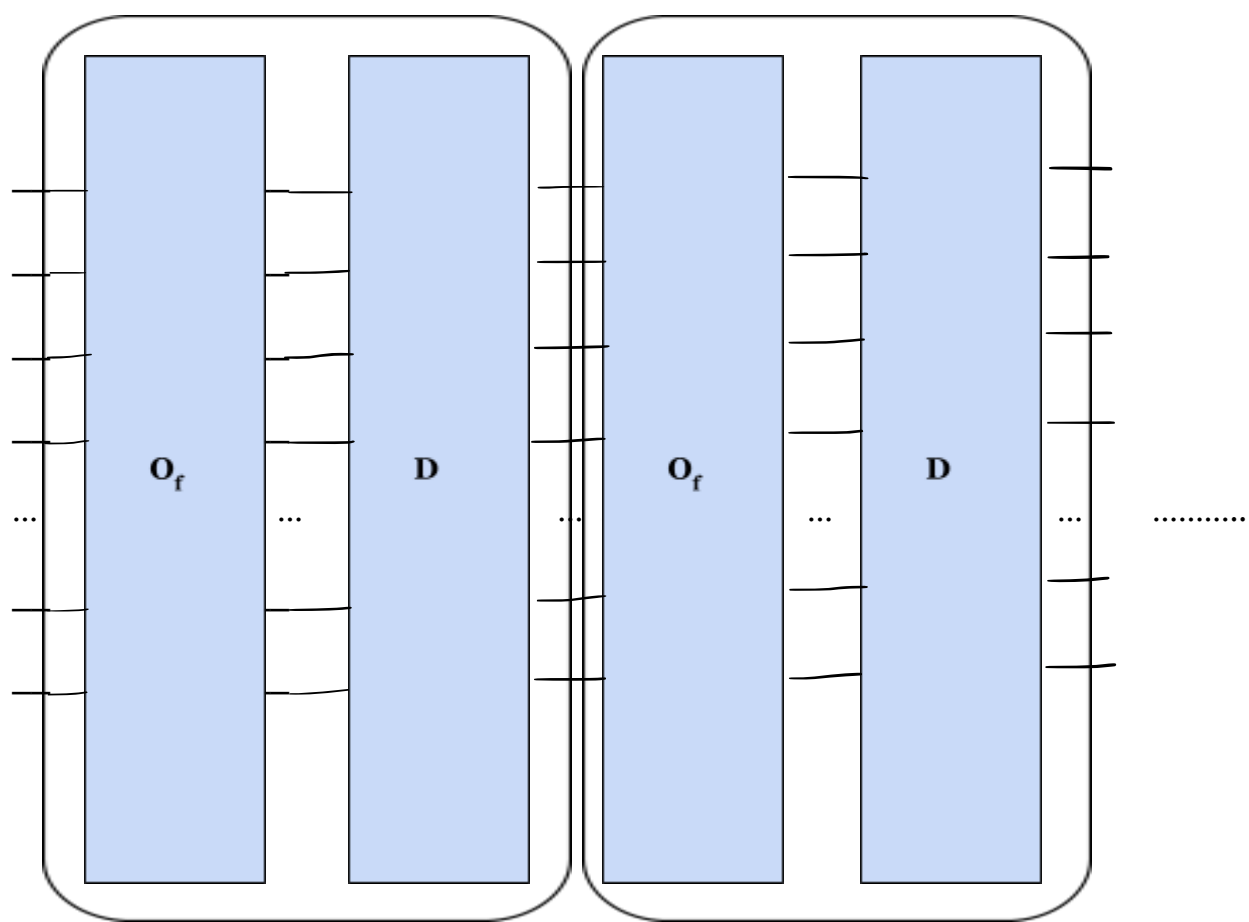
Note that N is the number of possible inputs and $N = 2^n$ where n is the number of qubits
We also assume $N \geq 4$

We have the Oracle $\mathbf{O}_f|x\rangle = -|x\rangle$ if $x = x^*$
 $= |x\rangle$ otherwise

Next, we have the Diffusion gate $\mathbf{D}$ where $D(\sum_t x_t |x\rangle) = \sum_t (2\mu - x_t) |x\rangle$ [μ is the average of all amplitudes]

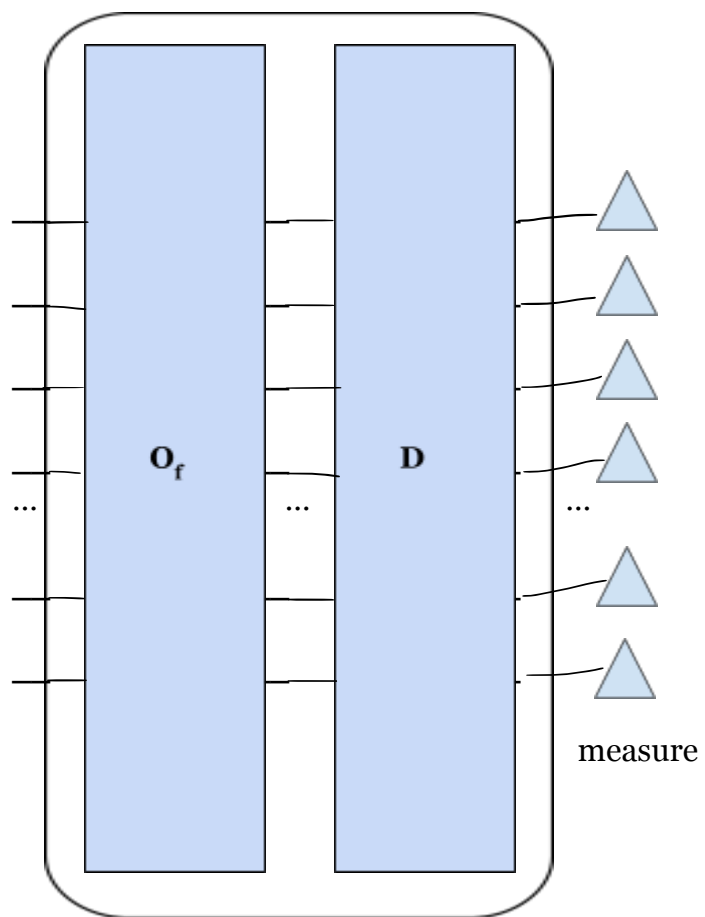
Constructing the circuit for Grover's Search we have,





Grover Iterate #t-1

Grover Iterate #t



Grover Iterate # $\frac{\sqrt{N}}{8}$

From Grover's circuit, we have a few definitions:

We define $\alpha^{(t)}$ as the amplitude of the marked state x^* after Iterate #t

We define $\beta^{(t)}$ as the amplitude of all other states ($x \neq x^*$) after Iterate #t

We have the values $\alpha_o = \frac{1}{\sqrt{N}} |x^* \rangle$, $\beta_o = \sum_x \frac{1}{\sqrt{N}} |x \rangle$ where $x \neq x^*$ and

$$\sum_{x \in \{0,1,\dots,N-1\}} \frac{1}{\sqrt{N}} |x \rangle = \frac{1}{\sqrt{N}} |x^* \rangle + \sum_x \frac{1}{\sqrt{N}} |x \rangle \text{ where } x \neq x^*$$

Next, we define $\mu^{(t)}$ as the average of all amplitudes after O_f gate in Iterate #t

$$\mu^{(t)} = \frac{-\alpha^{(t)} + (N-1)\beta^{(t)}}{N}$$

We define $\alpha^{(t+1)}$ in Iterate #t as

$$\begin{aligned} \alpha^{(t+1)} &= 2\mu^{(t)} - (-\alpha^{(t)}) \\ &= 2\mu^{(t)} + \alpha^{(t)} \end{aligned}$$

We define $\beta^{(t+1)}$ in Iterate #t as

$$\beta^{(t+1)} = 2\mu^{(t)} - \beta^{(t)}$$

Claim 1: If $\alpha^{(t)}$ is not too large, then $\alpha^{(t+1)}$ is greater than $\alpha^{(t)}$ by $O(\frac{1}{\sqrt{N}})$

$$\text{If } \alpha^{(t)} \leq \frac{1}{2}, \text{ then } \alpha^{(t+1)} \geq \alpha^{(t)} + \frac{1}{\sqrt{N}}$$

Proof: Using normalization:

$$(\alpha^{(t)})^2 + (N-1)(\beta^{(t)})^2 = 1$$

$$\leq \frac{1}{4} + (N-1)(\beta^{(t)})^2 = 1 \text{ [Since } \alpha^{(t)} \leq \frac{1}{2}, (\alpha^{(t)})^2 \leq \frac{1}{4}]$$

$$\Rightarrow 1 - \frac{1}{4} \leq (N-1)(\beta^{(t)})^2 \text{ [note that } \Rightarrow \text{ mean implies]}$$

$$\Rightarrow \frac{3}{4} \leq (N-1)(\beta^{(t)})^2$$

$$\Rightarrow (\beta^{(t)})^2 \geq \frac{3}{4(N-1)}$$

$$\Rightarrow (\beta^{(t)}) \geq \sqrt{\frac{3}{4(N-1)}} \rightarrow (A)$$

Now we have,

$$\mu^{(t)} = \frac{-\alpha(t) + (N-1)(\beta(t))}{N}$$

$$\geq \frac{-\frac{1}{2} + (N-1)\sqrt{\frac{3}{4(N-1)}}}{N} \quad \text{From equation (A)}$$

$$= \frac{-\frac{1}{2} + \frac{N-1}{2}\sqrt{\frac{3}{(N-1)}}}{N}$$

$$= \frac{\frac{1}{2}[(-1) + (N-1)\sqrt{\frac{3}{(N-1)}}]}{N}$$

$$= \frac{\frac{1}{2}[(-1) + \sqrt{(N-1)}\sqrt{3}]}{N}$$

$$= \frac{\frac{1}{2}[(-1) + \sqrt{3(N-1)}}}{N}$$

$$= \frac{1}{2} \frac{[(-1) + \sqrt{3(N-1)}}{N} \rightarrow (X)$$

We assumed $N \geq 4$, therefore, we want to check if $\frac{[(-1) + \sqrt{3(N-1)}}{N} \geq \frac{1}{\sqrt{N}}$ if $N \geq 4$

We have,

$$\frac{[(-1) + \sqrt{3(N-1)}}{N} \geq \frac{1}{\sqrt{N}}$$

$$(-1) + \sqrt{3(N-1)} \geq \frac{N}{\sqrt{N}}$$

$$(-1) + \sqrt{3(N-1)} \geq \sqrt{N}$$

Substituting $N = 4$, we get

$$\begin{aligned} (-1) + \sqrt{3(4-1)} &\geq \sqrt{4} \\ \sqrt{3(4-1)} &\geq 2 - 1 \end{aligned}$$

$$\sqrt{3(4-1)} \geq 3$$

$$\sqrt{9} \geq 3$$

$$3 \geq 3$$

$$3=3$$

$$\text{This means } \frac{[(-1) + \sqrt{3(N-1)}]}{N} \geq \frac{1}{\sqrt{N}}$$

$$\text{Therefore, equation (X) becomes } \frac{1}{2} \frac{[(-1) + \sqrt{3(N-1)}]}{N} \geq \frac{1}{2} \frac{1}{\sqrt{N}}$$

So we have

$$\begin{aligned} \alpha^{(t+1)} &= 2\mu^{(t)} + \alpha^{(t)} [\text{From our definition}] \\ &\geq 2 \frac{1}{2\sqrt{N}} + \alpha^{(t)} \\ &\geq \frac{1}{\sqrt{N}} + \alpha^{(t)} \\ &= \alpha^{(t)} + \frac{1}{\sqrt{N}} \end{aligned}$$

Hence Proved.

[Note that $\alpha^{(t)}$ should increase but in a controlled way]

$$\textbf{Claim 2: } \alpha^{(t+1)} \leq \alpha^{(t)} + \frac{2}{\sqrt{N}}$$

Proof: We know that

$$\mu^{(t)} = \frac{-\alpha^{(t)} + (N-1)(\beta^{(t)})}{N} \leq \frac{(N-1)(\beta^{(t)})}{N} \rightarrow A$$

[Since we know $\alpha^{(t)}$ is always positive $\Rightarrow \alpha^{(t)} \geq 0 \Rightarrow -\alpha^{(t)} \leq 0$]

From the normalization condition of a quantum state, we know that

$$\begin{aligned} (\alpha^{(t)})^2 + (N-1)(\beta^{(t)})^2 &= 1 \\ \Rightarrow (N-1)(\beta^{(t)})^2 &= 1 - (\alpha^{(t)})^2 \end{aligned}$$

$$\text{Therefore, } (N-1)(\beta^{(t)})^2 \leq 1$$

$$\begin{aligned} \Rightarrow (\beta^{(t)})^2 &\leq \frac{1}{N-1} \\ \Rightarrow (\beta^{(t)}) &\leq \frac{1}{\sqrt{N-1}} \rightarrow \mathbf{B} \end{aligned}$$

Therefore, now to prove the claim, we have

$$\begin{aligned} \alpha^{(t+1)} &= 2\mu^{(t)} + \alpha^{(t)} \\ &\leq 2\left(\frac{(N-1)(\beta^{(t)})}{N}\right) + \alpha^{(t)} \text{ [From equation A]} \\ &= 2 \frac{(N-1)}{N} (\beta^{(t)}) + \alpha^{(t)} \\ &\leq 2 \frac{(N-1)}{N} \frac{1}{\sqrt{N-1}} + \alpha^{(t)} \text{ [From equation B]} \\ &= 2 \frac{\sqrt{N-1}}{N} + \alpha^{(t)} \\ &< 2 \frac{\sqrt{N}}{N} + \alpha^{(t)} \\ &= \frac{2}{\sqrt{N}} + \alpha^{(t)} \\ &= \alpha^{(t)} + \frac{2}{\sqrt{N}} \end{aligned}$$

Hence proved.

Claim 3: We observe x^* with probability ≥ 0.1 i.e. show $(\alpha \frac{\sqrt{N}}{8})^2 \geq 0.1$ i.e. $\alpha \frac{\sqrt{N}}{8} \geq \sqrt{0.01} = 0.1$

Proof: If we assume $N \leq 16$, then $\alpha^{(0)} = \frac{1}{\sqrt{N}} = \frac{1}{4} > 0.1$

If we assume $N > 16$, then as long as $t \leq \frac{\sqrt{N}}{8}$,

$$\alpha^{(t)} \leq \frac{2}{\sqrt{N}} + \alpha^{(t-1)}$$

$$\leq \frac{4}{\sqrt{N}} + \alpha^{(t-2)}$$

$$\leq \frac{6}{\sqrt{N}} + \alpha^{(t-3)}$$

$$\leq \alpha^{(0)} + \frac{2}{\sqrt{N}} t \quad [t \leq \frac{\sqrt{N}}{8}]$$

$$\leq \frac{1}{\sqrt{N}} + \frac{2}{\sqrt{N}} \frac{\sqrt{N}}{8}$$

$$= \frac{1}{\sqrt{N}} + \frac{1}{4}$$

$$\leq \frac{1}{2} \quad \text{Since } N > 16$$

Now, since $\alpha^{(\frac{\sqrt{N}}{8})} \leq \frac{1}{2}$, then we have

$$\alpha^{(\frac{\sqrt{N}}{8})} \geq \frac{1}{\sqrt{N}} + \alpha^{\frac{\sqrt{N}}{8} - 1}$$

$$\geq \frac{2}{\sqrt{N}} + \alpha^{\frac{\sqrt{N}}{8} - 2}$$

$$\geq \frac{3}{\sqrt{N}} + \alpha^{\frac{\sqrt{N}}{8} - 3}$$

$$>= \frac{\sqrt{N}}{8} \frac{1}{\sqrt{N}} + \alpha^{(o)} \text{ [Iteratively applying]}$$

$$= \frac{1}{8} + \frac{1}{\sqrt{N}} > \mathbf{0.1}$$

Hence Proved.